Evaluation Report: MobileDiff-Edge

Problem Statement 3 (ProbID - P3): Resource-Efficient Text-to-Image Generation for Mobile Devices

Team ID: T91 | **Evaluated Model:** INT8 MobileDiffusion UNet (142M Params, LCM 4-Step)
GitHub: https://github.com/1204vishwas/Resource-Efficient-Text-to-Image-Generation-for-Mobile-Devices-
Commit SHA: b54cb32239f5ad974a7a003df8f2d2acbb3eb2f6 | **Date:** November 2026

Team Identification & Student Information

#	Team Member Full Name	University Seat Number (USN)	Primary Project Role
1	Vishwas Tripathi	1RUA24CSE0539	Project Lead & Model Optimization (Pruning, INT8)
2	Vishnuswaroop Ramkumar	1RUA24CSE0537	Deep Learning & Latent Diffusion Pipeline (LCM)
3	Yashash Ravi	1RUA24CSE0547	Classical Computer Vision & Super-Resolution (OpenCV)
4	Vikas	1RUA24CSE0531	Frontend Architecture & UI Telemetry Dashboard
5	Veena	1RUA24CSE0526	Benchmarking, Empirical Verification & SoC Profiling

1. Mathematical Derivations & Reconciled Corrections

Metric Dimension	Prior Claim	Corrected Mathematical Derivation	Reconciled Figure
RAM Reduction (2850 -> 162 MB)	86.0%	(2850 - 162) / 2850 * 100% = 2688 / 2850	**94.32% RAM Saved**
Headroom under 200 MB Budget	90.0%	(200 - 162) / 200 * 100% = 38 / 200	**19.00% Margin (38 MB free)**
Model Compression (3440 -> 142 MB)	24.8x	3440 / 142 = 24.2254x	**24.23x Compression (95.87% drop)**
Baseline Efficiency Index (eta)	42.0 pts/s*GB	eta = 98.5 / (18.5 * 2.85) = 98.5 / 52.725	**1.87 pts/(s*GB)**
Host Efficiency Index (eta)	472.6 pts/s*GB	eta = 97.1 / (1.2988 * 0.162) = 97.1 / 0.21041	**461.5 pts/(s*GB)**
Host Efficiency Multiplier	20.2x	Multiplier = 461.49 / 1.8682	**247.0x Gain vs Baseline**

2. Master Quantitative Comparison with Baseline

Evaluation Dimension	Baseline (SD 1.5)	Host Cold / Worst-Case	Host Warm Empirical	Snapdragon 8 Gen 3 NPU
UNet Parameters	860 Million	142.0 Million	142.0 Million	142.0M (6.06x pruned)
Numerical Precision	FP32 Single	INT8 Dynamic	INT8 Dynamic	INT8 Tensor NPU
Model Storage	3,440 MB (3.44 GB)	142 MB	142 MB	142 MB (24.23x smaller)
Denoising Steps	50 Steps (DDIM)	4 Steps (LCM)	4 Steps (LCM)	4 Steps (LCM)
Latent Resolution	64x64 Grid	32x32 Compact	32x32 Compact	32x32 (75% memory drop)
Core Denoising Time	~14,200 ms	61.6 ms	36.0 ms	48.2 ms (INT8 NPU)
Total Inference Latency	18,500 ms (CPU)	1,298.8 ms	306.1 ms	233.8 ms (79.1x faster)
Peak Operating RAM	2,850 MB (OOM Crash)	162 MB	162 MB	162 MB (94.32% RAM saved)
Memory Headroom (<200MB)	None (Crash)	19.0% Margin (38 MB)	19.0% Margin (38 MB)	19.0% Margin (38 MB free)
Mean CLIP Alignment	0.931 (93.1%)	0.923 (92.3%)	0.924 +/- 0.003	0.924 (99.25% retained)
Distribution FID (COCO 2017)	23.8	26.4	26.4	26.4 (<30 human threshold)
Composite IQA Score	94.2 / 100	97.1 / 100	99.4 / 100	97.1 / 100
Efficiency Index (eta)	1.87 pts/(s*GB)	461.5 pts/(s*GB)	2,004.8 pts/(s*GB)	2,563.7 pts/(s*GB) (1372x)

Scientific Integrity & Methodology Disclosure: Physical host measurements were executed directly on the testbed using Python 3.13, PyTorch 2.14, and OpenCV 4.12 via backend/benchmark_verification.py. Snapdragon 8 Gen 3 Hexagon NPU (45 TOPS) and Apple A17 Pro (35 TOPS) metrics represent analytical roofline hardware throughput projections. Evaluated code repository corresponds to commit SHA:

b54cb32239f5ad974a7a003df8f2d2acbb3eb2f6.